CS311 Operating Systems
Assignment 03 - xv6 Page Tables and Traps
Due : Task 1: 23h55 Sunday 15th November, 2026
Task 2: 23h55 Sunday 22 November, 2026.

FCS&E
GIKI

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1 Objectives

(a) Learn about virtual memory management in xv6.

(b) Learn about traps in xv6.

(c) Test the student's ability do error handling in code.

(d) Test the student's ability to follow written instructions.

2 Description

In this assignment you will be working with xv6. You'll learn how xv6 compiles the kernel and user space programs. How to add a new user space program to xv6. You'll also learn about how system calls in xv6 work, how to use them, and how to add a new system call to the xv6 kernel which takes some arguments from the user and performs some action on their behalf.

The evaluation based on this assignment will weigh 5% of your course grade.

Note: Please start early so you have time to learn things which you don't.

3 Tasks

3.1 Task 0

Install [1] xv6 and qemu locally on your system. Compile and run xv6.

3.2 Task 1

Do the third xv6 lab (on Page Tables) [2] implementing all the 5 tasks: `inspect`, `walk`, `detect`, `speed up pid`, `superpages`.

[1] https://pdos.csail.mit.edu/6.828/2026/tools.html
[2] https://pdos.csail.mit.edu/6.828/2026/labs/pgtbl.html

3.3 Task 2

Do the fourth xv6 lab (on Traps) [3] implementing all the listed tasks: `risc-v assembly, backtrace, alarm`.

4 Resrouces

- xv6 [4] is an educational operating system developed at MIT and used by many other universities to let their students practice Operating Systems concepts.
- The information is available on MIT website [5].
- The latest version is written for RISC-V architecture and the source code has been made available on github. [6]
- There's a companion booklet that explains the overall architecture and some details of the code. [7] The book repository, when compiled, also compiles the source code into a PDF booklet annotated line numbers for each line of code. The code book often refers to these source code line numbers so it's good if you have then both compiled to PDFs.
- There's also this youtube video playlist where a professor explains the workings of each line of xv6 source code in detail. [8] Very useful.

5 Error Checking

- When working with System calls, it is extremely important that you always check the return values of the system calls to verify whether the call was successful or not. The function documentation would usually contain all the information you will need. You should always read the function documentation. From now on, start reading xv6 codes too.
- You would lose marks if your program crashes during use. Use `gdb` to find any bug in your program.
- It is the programmer's responsibility to free any system resources i.e. memory , file descriptors, etc., they have acquired from the system. Your program should always free system resources once it is done using them. `Valgrind` is a tool that you can run to check for memory leaks, etc.
- Code should be properly indented, readable and commented.
- Use xv6 Makefile to compile your codes. Try reading through it and understand what options, etc., it user for compiler and linker.

[3] https://pdos.csail.mit.edu/6.828/2026/labs/traps.html
[4] https://en.wikipedia.org/wiki/Xv6
[5] https://pdos.csail.mit.edu/6.828/2026/xv6.html
[6] https://github.com/mit-pdos/xv6-riscv
[7] https://github.com/mit-pdos/xv6-riscv-book.git
[8] https://www.youtube.com/watch?v=xieHeMeu1HQlist=PLbtzT1TYeoMj$_Y$$vECDl0YqipVZqucZHXs$

6 Submission Instructions

1. Use the xv6 repository you forked into your github account for last assignment.
2. Create a separate branch for assignment 3 and do all the work on it. Do no modifications to this branch after the deadline.

7 Rubric

Category	Marks
Followed submission insns	10 marks
Code was readable + Compiled without warnings + Program handles errors well	20 marks
Task1 working properly	30 marks
Task2 working properly	30 marks
Total	100 marks